

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

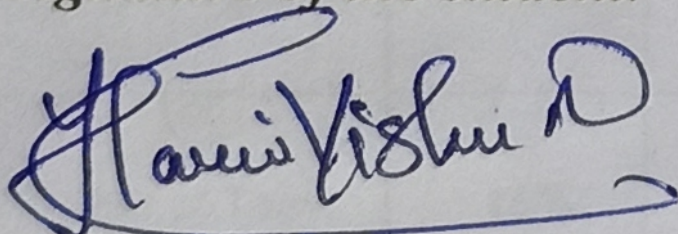
INDUSTRY PROBLEM SOLVING TASK

Code of Conduct:

I HARI VISHNU N / 192524319 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

INDUSTRY PROBLEM SOLVING TASK

S.No	Industry Problem Solving Task Questions	Marks
1	A cloud storage company is facing performance issues due to frequent file access delays. Analyze whether contiguous, linked, or indexed allocation would be most suitable for improving access speed and justify your choice with reasons.	5
2	A data center experiences high disk head movement while processing user requests. Suggest the most appropriate disk scheduling algorithm (FCFS, SSTF, SCAN) to reduce seek time and explain how it improves performance in this scenario.	5
3	An operating system uses a bitmap for free space management but struggles with slow allocation when the disk size increases. Propose an optimization technique to improve the efficiency of bitmap-based allocation and justify your solution.	5
4	A real-time application requires consistent and predictable disk access times. Evaluate whether SCAN or C-SCAN would be more suitable and explain your reasoning based on system requirements.	5
5	A company wants to design a hybrid storage system combining file allocation and disk scheduling techniques. Suggest a suitable combination (allocation+scheduling) for handling large-scale data access efficiently and justify your choice based on performance factors.	5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

1) Best file Allocation Method:-

Contiguous Allocation is most suitable when the main requirement is fast file access.

Reason:-

File blocks are stored in consecutive disk location.
Provides very fast sequential and direct access.

Requires fewer disk head movement.

Simple to manage and access.

Suitable for frequently accessed large files.

Limitation: External fragmentation can occur, and expanding a file can be difficult.

Conclusion: For a cloud storage system where access speed is the priority, contiguous allocation is preferred.

2) Best Disk Scheduling Algorithm:

SSTF (Shortest seek time first) is suitable for reducing disk head movement.

Reason:

Selects the request closest to the current head position.

Reduces the distance travelled by the disk head.

Provides lower average seek time compared with FCFS.

Improves disk throughput and response time.

Limitation :- Requests far from the current head may experience starvation.

Conclusion :- SSTF is a good choice when the primary objective is minimizing seek time.

3) **Optimizing Bitmap Allocation** :-

A suitable Bitmap opt. is to maintain a free block pointer/index along with the bitmap.

Reason :-

Reduces unnecessary scanning.

Allocation becomes faster as disk size increases.

The bitmap can also be divided into chunks and searched word by word for free bfb.

Conclusion :- Using free space pointer plus word level chunk-based searching significantly improves bitmap allocation.

4) **SCAN VS C-SCAN** :-

For a real-time application requiring consistent and predictable access time, C-SCAN is more suitable.

Reason :-

C-SCAN request in one direction only.

After reaching one end, the head returns to the beginning and continues servicing request.

Provides more uniform waiting time.

SCAN can provide good performance but request near the middle may receive better service than request near end.

Conclusion:- C-SCAN is preferred when fairness and predicted response time are most important than minimum total head movement.

5) Hybrid storage system:-

A suitable combination is;

Contiguous allocation + SSTF scheduling.

Reason:-

Contiguous allocation provides fast direct and sequential file allocation.

SSTF select nearest pending disk.

Together, they reduce both file access and disk seek time for improvement.

Suitable for large scale

Improves overall throughput and time.

Limitation: contiguous Allocation May suffer from external fragmentation while SSTF can use starvation.

Conclusion: contiguous + SSTF provides high performance when speed and low seek time have priority.